



Bruna Strack Cândido

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● ABOUT ME

I am a data scientist with a background in oceanography and a focus on turning complex Earth system data into reliable forecasts. My path started at Universidade Federal de Santa Catarina, where studying the ocean taught me to work with large, noisy, physically constrained datasets. From there I spent nearly five years at Fractal Engenharia, building and operating real-time flood forecasting systems that ingested NWP data (GFS, ECMWF, CFS, WRF) and weather radar, serving the Brazilian Civil Defence and critical infrastructure operators across Brazil. During several extreme flood events in the South of Brazil, those forecasts helped save lives, which made me feel really proud of this work. Wanting to deepen the science behind these systems, I joined the Erasmus Mundus Master's Copernicus in Digital Earth, specialising in GeoData Science, at Université Bretagne Sud and Paris-Lodron University of Salzburg. I am now developing my thesis at ESA's Φ -lab, applying Physics-Informed Neural Networks, constrained by the governing equations of ocean physics, to reconstruct subsurface variables such as temperature, salinity, mesoscale eddies and currents from satellite and Argo data. What unites all of this is a drive to make scientific models that actually perform when it matters most.

● WORK EXPERIENCE

VISITING RESEARCHER – EUROPEAN SPACE AGENCY - ESA – 04/05/2026 – 31/07/2026 – FRASCATI, ITALY

Visiting Researcher at the ESA's Φ -lab, working on my Master's Thesis on Physics-Informed Neural Networks (PINNs) for reconstructing ocean variables.

Main activities:

- Developed deep learning models for ocean data reconstruction using satellite altimetry and Argo float data.
- The approach embeds governing ocean physics equations (momentum, continuity, thermodynamic) directly into the neural network loss function, enabling physically consistent reconstructions even in data-sparse regions, with direct relevance to marine submersion and extreme coastal event modelling.
- Building end-to-end ML pipelines in Python and PyTorch for large-scale oceanographic and Earth observation datasets.

RESEARCH INTERN – CMCC FOUNDATION - CENTRO EURO MEDITERRANEO SUI CAMBIAMENTI CLIMATICI – 01/07/2025 – 31/08/2025 – LECCE, ITALY

- Research within the Maritime Solutions / Global Coastal Ocean (GOCO) unit on oil spill segmentation in synthetic aperture radar (SAR) imagery using deep learning.
- Created a Python pipeline to evaluate and compare multiple deep learning models for oil spill delineation in SAR imagery, with the goal of automating slick detection as input to the MEDSLIK-II oil spill simulation and transport model.
- Processed SAR images and datasets and ran experiments to assess model performance across real maritime scenarios.

HYDROLOGY AND SOFTWARE DEVELOPER – FRACTAL ENGENHARIA E SISTEMAS – 01/11/2019 – 31/08/2024 – FLORIANÓPOLIS, BRAZIL

- Calibrated and operated real-time hydrological forecast models (SMAP, SPEHC, PMTankModel) for river basins across Brazil, ingesting NWP outputs from GFS, ECMWF, CFS and WRF models, as well as weather radar data from the Santa Catarina State Civil Defence.
- Collaborated closely with Civil Defence meteorologists and hydrology teams to calibrate alert thresholds and validate forecast outputs in operational conditions.
- Forecasts were operationally used by the Brazilian Civil Defence for flood early warning and emergency response. This system directly contributed to saving lives during several extreme flood events in southern Brazil.

- Developed forecast and management services for hydroelectric power plant operators and electric energy companies, enabling clients to manage reservoir operations and optimise energy generation scheduling based on inflow predictions, delivering direct economic value from hazard intelligence.
- Contributed to the development of the SIG²A platform (Integrated Asset Management System): a web GIS solution concentrating dam safety analysis tools, hydrological situation dashboards and real-time forecast outputs for civil infrastructure and energy sector clients.
- Delimited hydrographic basins and processed spatial data in QGIS and ArcGIS; statistical analysis and process automation in Python and Matlab.
- Stack: Python, Matlab, JavaScript, AngularJS, ReactJS, PostgreSQL/PostGIS, GIS, SCRUM.

UNDERGRADUATE RESEARCHER – COASTAL OCEANOGRAPHY LABORATORY (FEDERAL UNIVERSITY OF SANTA CATARINA) – 01/08/2021 – 01/08/2022 – FLORIANÓPOLIS, BRAZIL

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- Developed a bathymetric restitution model for sandy beaches from Sentinel-2 satellite imagery, directly relevant to coastal flood risk and marine submersion depth mapping.
 - Analytical bathymetry inversion provides a low-cost method for nearshore depth estimation ahead of storm surge and submersion events.
 - Work published at Coastal Sediments 2023 international conference.

UNDERGRADUATE RESEARCHER – DYNAMIC OCEANOGRAPHY LABORATORY - LABDINO - (FEDERAL UNIVERSITY OF SANTA CATARINA) – 01/08/2019 – 28/02/2020 – FLORIANÓPOLIS, BRAZIL

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- Statistical validation of wave height, period and direction from numerical ocean models using MatLab, building expertise in ocean dynamics relevant to marine hazard assessment.

CO-FOUNDER AND MEMBER OF THE EXECUTIVE COMMITTEE – GARAPUVU – 01/08/2017 – 01/07/2020 – FLORIANÓPOLIS, BRAZIL

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- Co-founded a digital platform connecting environmentally conscious citizens with sustainable businesses across the Florianópolis metropolitan region.
 - Selected for the CocreationLab business pre-incubation programme; I was responsible for establishing internal processes and coordinating team operations across working groups.

VOLUNTEER – TÉTIS, JUNIOR OCEANOGRAPHY COMPANY - UFSC – 01/04/2017 – 01/10/2019 – FLORIANÓPOLIS, BRAZIL

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- Contributed to environmental conservation projects and organisation of academic events, including beach cleanup initiatives, within a student-led multidisciplinary team.

● **EDUCATION AND TRAINING**

30/09/2024 – 30/09/2026 Vannes, France

MASTER IN DATA SCIENCES PARCOURS GEODATA SCIENCE Université Bretagne Sud and Paris Lodron University of Salzburg

Currently studying. Thesis: Physics-Informed Neural Networks for subsurface ocean reconstruction (ESA Φ-lab, Frascati, Italy). EQF Level 7.

Website <https://master-cde.eu/> | Level in EQF EQF level 7

15/02/2017 – 05/10/2022 Florianópolis, Brazil

BACHELOR'S DEGREE IN OCEANOGRAPHY Federal University of Santa Catarina (UFSC)

Website ufsc.br

● **LANGUAGE SKILLS**

Mother tongue(s): **PORTUGUESE**

Other language(s):

	UNDERSTANDING		SPEAKING		WRITING
	Listening	Reading	Spoken production	Spoken interaction	
ENGLISH	C1	C1	C1	C1	C1
SPANISH	B1	B1	B1	B1	B1
FRENCH	B1	C1	A2	A2	A2

Levels: A1 and A2: Basic user; B1 and B2: Independent user; C1 and C2: Proficient user

SKILLS

GIT version control, Linux Command | programming: Python, MATLAB and SQL | JavaScript, HTML, CSS, AngularJS, ReactJS | GIT, Bitbucket, PostgreSQL | Data formats: csv, JSON, NetCDF | Statistics Programme R | Unix Shell (SSH) | PostgreSQL (+PostGIS) | GIS: QGis, ArcGis and eCognition | Satellite Data Process | PyTorch | Efficient Remote Sensing | Big Data | Machine Learning, Deep Learning, PyTorch

PUBLICATIONS

2023
[ANALYTICAL APPROACH FOR BATHYMETRY INVERSION FROM SENTINEL-2 OR UAV IMAGERY](#)

CANDIDO, BS, FONTOURA KLEIN, AHD, ALMEIDA, LP, PEREIRA, P., & LEAL-ALVES, DC (2023). **ANALYTICAL APPROACH FOR BATHYMETRY INVERSION FROM SENTINEL-2 OR UAV IMAGERY**. In Coastal Sediments 2023: The Proceedings of the Coastal Sediments 2023 (pp. 1530-1542).

CANDIDO, BS, FONTOURA KLEIN, AHD, ALMEIDA, LP, PEREIRA, P., & LEAL-ALVES, DC (2023)

2022
[Reconstituição do perfil batimétrico de praias a partir de imagens do satélite Sentinel-2](#)

Cândido, BS (2022).